Lecture 06: Structures Miscellaneous

Today’s Lecture

- Passing Structure Members as arguments to Function
- Passing Structure Variables as Parameters
- Returning Structure from Function
- Pointers to structure variables
- Passing Structure Pointers as Argument to a Function
- Returning a Structure Pointer from Function
- Passing Array of structures
- Dynamic Memory Allocation (DMA) of Structure Type Variables

Example : Passing a Structure Member

- We can pass individual members to a function just like ordinary variables.

```
struct student {
    string name;
    int roll_no;
    int marks;
};
void print_struct(string name, int roll_no, int marks);
int main() {
struct student stu = {“Rashid", 1, 78};
print_struct(stu.name, stu.roll_no, stu.marks);
    return 0;
}
void print_struct(string name, int roll_no, int marks) {
 cout<<"Name: "<< name<<endl;
  cout<<"Roll no: "<< roll_no<<endl;
  cout<<"Marks: "<< marks<<endl;
}
```

```
Name: Rashid
Roll no: 1
Marks: 78
```

Passing Structure Variables as Parameters

- An individual structure member may be passed as a parameter to a function as shown in the last example.
- However, passing individual structure members, if there are many, is not only tiresome but also error-prone.
- So in such cases instead of passing members individually, we can pass structure variable itself.

Example : Passing a Structure as argument

- We can pass structure variable to a function just like ordinary variables.

```
struct student
{     char name[20];
      int roll_no, marks;
};
void print_struct( student stu); //prototype of function
int main() {
    struct student stu = {“Rashid", 1, 78}; //aggregate initialization
    print_struct(stu);
    return 0;
}
void print_struct(student stu) {
    cout<<"Name: "<< stu.name;
  cout<<"Roll no: "<< stu.roll_no;
  cout<<"Marks: "<< stu.marks;
}
```

```
Name: Rashid
Roll no: 1
Marks: 78
```

Returning Structure from Function

- Just as we can return fundamental types and arrays, we can also return a structure from a function.
- To return a structure from a function we must specify the appropriate return type in the function definition and declaration. For example:

```
student change(struct student);
```

- This function accepts an argument of type struct student and returns an argument of type struct student.

Example : Returning Structure from Function

```
struct student {
    char name[20];
    int roll_no, marks;
};
void print_struct(struct student);          //prototype of function
student change(struct student);             //prototype of function
int main() {
    struct student stu = {“Sohail", 1, 55};
    print_struct(stu);
   stu = change(stu);
   cout<<"After calling change() function"<<endl;
   print_struct(stu);
    return 0;
}
void print_struct(struct student stu) {
    cout<<"Name: "<< stu.name<<endl<<"Roll no: “<<endl<< stu.roll_no<<"Marks: "<< stu.marks;
}
struct student change(student csstu) {
csstu.roll_no  = 21;
 return csstu;   }
```

```
Name: Sohail
Roll no: 1
Marks: 55

After calling change() function
Name: Sohail
Roll no: 21
Marks: 55
```

Passing Structure Variables as Parameters

- Unlike arrays, the name of structure variable is not a pointer.
- So when we pass a structure variable to a function, the formal argument of print_struct() is assigned a copy of the original structure.
- Both structures reside in different memory locations and hence they are completely independent of each other.
- Any changes made by function print_struct() doesn't affect the original structure variable in the main() function.

Pointers to structure variables

- Pointers of structure variables can be declared like pointers to any basic data type

```
Student csstudent, *ptrstudent;
ptrstudent = &csstudent;
```

- To access the member of the struct using pointer to struct, we can use (.) operator by first dereferencing the pointer, for example

```
(*ptrstudent).roll_number;
```

- Alternatively, members of a pointer structure type variable can be accessed using arrow (->) operator

```
ptrstudent-> roll_number=20; //alternate: (*ptrstudent).roll_number;
strcpy( ptrstudent->Name, “Ali”);
```

- Members of the structure variables can be accessed directly or indirectly via pointer.

```
cout<<"Name: "<< csstudent.name<<endl<<ptrstudent->rollnumber;
cout<<"Roll no: "<< csstudent.roll_no<<endl<<ptrstudent->Name;
```

Mixing pointers and non-pointers to members

- The member selection operator is always applied to the currently selected variable.
- If you have a mix of ***pointers*** and ***normal member variables***, you can see member selections where . and -> are both used in sequence:

```
struct Employee {
 int employees;
};

struct Company{
  string name;
 Employee emp;
};
int main(){
Company IBM = { "IBM", { 5 } };
Company *ptr = &IBM ;     // ptr is a pointer, use ->
// emp is not a pointer, use .
 cout << (ptr->emp).employees << '\n';
return 0;
}
```

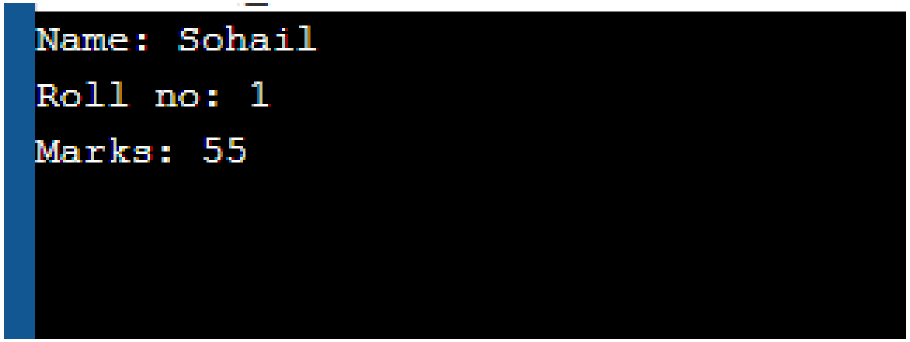
Passing Structure Pointers as Argument to a Function

Passing structure variables as pointers

- When we pass structure variables by value, a copy of the structure is passed to the formal argument.
- If the structure is large it can take quite a bit of time which make the program inefficient.
- Additional memory is required to save every copy of the structure.
- It is better to pass structure pointers as arguments to a function

Passing structure variables as pointers

```
struct student {
    char name[20];
    int roll_no;
    int marks;
};
void print_struct(student *);          //prototype of function
int main() {
    struct student stu = {“Sohail", 1, 55};
    print_struct(& stu);
    return 0;
}
void print_struct(student * csstu) {
    cout<<"Name: "<< csstu->name<<endl;         //alternate. cout<< (*csstu).name
   cout<<"Roll no: "<< csstu->roll_no<<endl;     //alternate. cout<< (*csstu).roll_no
   cout<<"Marks: "<< csstu->marks<<endl;         //alternate. cout<< (*csstu).marks
}
```



Passing structure variables as pointers

- Passing structure pointer to a variable lets the function modify the values of the members of the structure, which will also be reflected in the main.

Returning a Structure Pointer from Function

- A function can also return a pointer to structure variable.
- Appropriate return type shall be specified in the function definition and function declaration. Such as

```
struct student * change(struct student *);
```

Example : Returning a Structure Pointer from Function

```
struct student {
    string name;
    int roll_no, marks;
};
void print_struct(struct student stu *);  // function prototype
struct student * change(struct student *);        // function prototype
int main() {
    struct student stu = {“Sohail", 1, 55};
struct student *ptr_stu1 = &stu, *ptr_stu2;
    print_struct(ptr_stu1);
   ptr_stu2 = change(ptr_stu1);
   cout<<"After calling change() function"<<endl;
   print_struct(ptr_stu2);
    return 0;
}
void print_struct(struct student * stu) {
cout<<"Name: "<< stu->name<<endl<<"Roll no: “<<endl<< stu->roll_no<<"Marks: "<< stu->marks;
}
struct student * change(student *csstu) {
csstu->marks  = 50;
 return csstu;   }
```

```
Name: Sohail
Roll no: 1
Marks: 55

After calling change() function
Name: Sohail
Roll no: 1
Marks: 50
```

PASSING ARRAY OF STRUCTURE TO FUNCTION

Passing Array of Structures to Function

- Array of structures can be passed to functions the same way as array of integers is passed to function.
- For example, to pass an array of struct student, the prototype of function print_struct() is declared which accepts an argument of type array of structures.

void print_struct (struct student str_arr[], int size);

- In main(), an array of structures of type student can be declared as:

struct student stu[3] = { {“John", 1, 55 }, {“Tim", 2, 65 }, {“Green", 3,75}};

- Similarly, in main() print_struct() can be called as:

print_struct(stu,size);

- Note that the name of the array is a constant pointer to the 0th element of the array, the formal argument of print_struct() is assigned the address of variable students.

Passing Array of Structures to Function

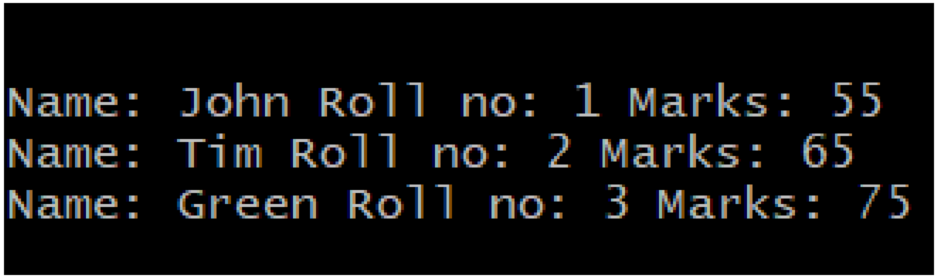
- Print_struct() function can then iterate through the elements of the array that is passed to it as argument. For example:

```
void print_struct(struct student stu[ ], int size) {

          int i;

             for(i= 0; i<size; i++)
          {
          cout<<"Name: "<< stu[i].name;
          cout<<"Roll no: "<< stu[i].roll_no;
          cout<<"Marks: "<< stu[i].marks;
          }
}
```

Complete Example: Passing Array of Structures to Function

```
struct student {
    string name;
    int roll_no, marks;
};
void print_struct (struct student []); //Function prototype
int main() {
struct student stu[3] = { {“John", 1, 55 },  {“Tim", 2, 65 }, {“Green", 3,75}};
print_struct(stu);
return 0;
}
```

```
void print_struct(struct student stu[ ]) {
   for(int i= 0; i<3; i++)  {
            cout<<"Name: "<< stu[i].name;
            cout<<"Roll no: "<< stu[i].roll_no;
            cout<<"Marks: "<< stu[i].marks;
            }
}
```



Dynamic Memory Allocation (DMA) of Structure Type Variables

- We can also dynamically allocate the memory of any structure type variable using new operator.
- For example.

```
Student *ptrstudent;
ptrstudent = new Student;
```

- Syntax is similar to dynamically allocating memory for variables of other normal data types such as int, float etc.

```
int *PtrToInt;
PtrToInt = new int;
```

- We can delete memory allocated at execution time using delete

```
delete ptrstudent;
```

Dynamic Memory Allocation (DMA) of Structure Type Variables

- The function named allocate() dynamically allocates memory for a struct student and returns the pointer pointing to the dynamic memory that can be used in the main() function. We can also dynamically allocate the memory of any structure type variable using new operator.

```
struct student {
    string name;
    int roll_no, marks;
};
student * allocate (); //Function
prototype
int main() {
student *sptr = allocate();
//sptr and then delete memory
return 0; }
student * allocate() {
student * ptr = new student;
return ptr; }
```

Dynamic Memory Allocation (DMA) of Structure Type Variables

- Here is an alternative approach:
- Why pass by reference:
  Passing by reference (student * &ptr) → Function modifies the actual pointer → Changes persist in main().

```
struct student {
    string name;
    int roll_no, marks;
};
void allocate(student *&ptr) {
ptr = new student;
return ptr; }
int main() {
student *sptr;
allocate(sptr); //nothing to return
//sptr and then delete memory
return 0; }
```